



# SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

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Module 12: Applications of Random Graphs

Lecture 01: Introduction to Network Medicine



# Why Network Medicine?

## Background

Traditional medicine often studies diseases by isolating single genes, proteins, or organs. This reductionist approach works well for monogenic diseases but fails to explain complex, chronic, and multifactorial disorders such as cancer, diabetes, cardiovascular diseases, and neurological conditions.

## Need for a New Paradigm

Most diseases arise from interactions among many biological components. Network medicine addresses this by studying how failures emerge from the interaction structure of biological systems rather than from isolated defects.

# Limitations of the One-Gene–One-Disease Model

## Genetic Complexity

Genome-wide association studies show that most diseases involve hundreds of genes, each contributing a small effect. No single mutation fully explains disease onset or progression.

## Clinical Consequences

Patients with the same diagnosis often respond differently to the same drug. Single-target therapies frequently fail due to compensatory pathways in the biological network.

# Core Idea of Network Medicine

## Systems Perspective

Network medicine views the human body as an interconnected system where biological function emerges from interactions among components across multiple scales.

## Disease Definition

Disease is defined as a perturbation of the normal network structure or dynamics, leading to system-level dysfunction rather than isolated molecular failure.

# Introduction to Network Theory

## Mathematical Representation

A biological system is represented as a graph:

$$G = (V, E)$$

where  $V$  represents biological entities (genes, proteins, metabolites) and  $E$  represents interactions between them.

## Why Graphs?

Graph theory allows complex biological interactions to be quantified, visualized, and analyzed using rigorous mathematical tools.

# Adjacency Matrix Representation

## Matrix Formulation

Networks are stored as adjacency matrices:

$$A_{ij} = \begin{cases} w_{ij} & \text{if nodes interact} \\ 0 & \text{otherwise} \end{cases}$$

where  $w_{ij}$  represents interaction strength.

## Disease Interpretation

Disease corresponds to systematic changes in this matrix, such as loss of edges, gain of abnormal interactions, or altered interaction weights.

# Types of Biological Networks

## Molecular Networks

Gene regulatory networks, protein–protein interaction networks, signaling pathways, and metabolic networks capture molecular-level interactions.

## Higher-Level Networks

Cell–cell communication networks, organ interaction networks, and disease networks link molecular changes to physiological outcomes.

# The Human Interactome

## Definition

The interactome is the complete map of molecular interactions occurring in human cells. It integrates protein–protein, regulatory, signaling, and metabolic interactions.

## Relevance to Disease

Disease-associated genes are not randomly distributed but cluster in specific regions of the interactome, forming disease-related subnetworks.



# Scale-Free Nature of Biological Networks

## Degree Distribution

Biological networks typically follow a power-law distribution:

$$P(k) \sim k^{-\gamma}$$

meaning few nodes (hubs) have many connections, while most have few.

## Biological Implications

This structure provides robustness against random failures but makes the system vulnerable to targeted attacks on hubs, which can trigger disease.

# Centrality Measures and Node Importance

## Quantifying Importance

Key metrics include degree centrality and betweenness centrality:

$$C_B(i) = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}}$$

which measures how often a node lies on shortest paths.

## Disease Relevance

Nodes with high centrality often regulate information flow. Their disruption can cause large-scale network dysfunction.

# Disease Modules

## Concept

A disease module is a connected subnetwork whose disruption leads to a specific disease. Different molecular defects within the same module can produce the same clinical phenotype.

## Mathematical Detection

Modules are identified using clustering algorithms and modularity optimization, quantified by the modularity score  $Q$ .

# Genetic Heterogeneity Explained

## Patient Variability

Patients with the same disease often have different genetic mutations. Network medicine explains this through shared disruption of the same disease module.

## Clinical Insight

This explains why genetic testing may identify different mutations yet lead to the same diagnosis and symptoms.

# Disease–Disease Relationships

## Network Proximity

Diseases are related if their modules are close in the interactome. The average shortest-path distance quantifies this relationship.

## Comorbidity

Diseases with overlapping or nearby modules tend to co-occur, explaining observed patterns of comorbidity in clinical practice.

# Disease Progression as Network Failure

## Percolation Theory

Disease progression can be modeled as node or edge removal. Targeted removal of hubs leads to rapid network fragmentation.

## Clinical Interpretation

This explains sudden worsening of disease after long stable periods when critical network thresholds are crossed.

# Drug Action as Network Perturbation

## Mathematical Model

Drug action modifies the network:

$$A' = A + \Delta A$$

where  $\Delta A$  represents inhibited or enhanced interactions.

## Therapeutic Goal

Effective drugs selectively normalize disease modules while preserving overall network stability.

# Drug Repurposing Using Networks

## Network Proximity Principle

If drug targets lie close to a disease module, the drug may be effective even for a new indication.

## Advantages

This approach reduces development cost, time, and risk compared to de novo drug discovery.



# Network-Based Biomarkers

## Beyond Single Molecules

Network biomarkers measure changes in connectivity, module activity, and network dynamics over time.

## Clinical Value

They improve early diagnosis, prognosis, and prediction of treatment response.

# Personalized Network Medicine

## Patient-Specific Networks

Each patient has a personalized network constructed from genomic, transcriptomic, proteomic, and clinical data.

## Precision Therapy

Treatment aims to restore the patient's network toward a healthy reference state rather than targeting single molecules.

# Role of Artificial Intelligence

## Machine Learning on Graphs

Graph neural networks learn patterns directly from network structure to predict disease risk and drug response.

## Clinical Decision Support

AI enhances diagnosis, prognosis, and personalized treatment planning in network medicine.

# Challenges and Limitations

## Data Challenges

Incomplete interactomes, noisy biological data, and limited longitudinal datasets remain major challenges.

## Clinical Translation

Integrating network models into routine clinical workflows requires validation, interpretability, and regulatory approval.

# Conclusion and Future Directions

## Paradigm Shift

Network medicine shifts healthcare from reductionist to systems-level understanding of disease.

## Future Outlook

Digital twins, real-time network monitoring, and preventive interventions based on early network changes represent the future of precision medicine.

## References

- 1 A.L. Barabási, N. Gulbahce & J. Loscalzo, Network medicine: a network-based approach to human disease. *Nat Rev Genet* 12, 56–68 (2011): <https://doi.org/10.1038/nrg2918>
- 2 A. Banerjee, R.P. Goswami, & M. Chatterjee, Network theoretic analysis of JAK/STAT pathway and extrapolation to drugs and viruses including COVID-19. *Sci Rep* 11, 2512 (2021): <https://doi.org/10.1038/s41598-021-82139>

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**Module 12: Applications of Random Graphs**

**Lecture 02: Selected Case Study on Network Medicine**





# Biological Motivation

## JAK/STAT Pathway

- JAK/STAT is a central signaling pathway regulating cytokine signaling, immune activation, and inflammation.
- Dysregulation causes autoimmune diseases such as rheumatoid arthritis and cytokine storms in viral infections.
- Many drugs and viruses act by inhibiting specific proteins in this pathway.

## Key Idea

- A signaling pathway is a network.
- Drug or viral action corresponds to deleting nodes from this network.
- The remaining network size reflects biological activity.

# Basic Graph Theory

## Graphs

- A graph is a pair  $G = (V, E)$  where  $V$  is a set of vertices and  $E \subseteq V \times V$  is a set of edges.
- If  $(u, v) \in E$  and  $(v, u) \in E$  are treated as distinct, the graph is directed.
- If edges have no direction, the graph is undirected.

## JAK/STAT as a Graph

- Vertices represent proteins.
- Directed edges represent signal flow between proteins.
- Removing a vertex corresponds to blocking a protein biologically.

# Subgraphs and Deletion

## Vertex Deletion

- If  $S \subseteq V$ , the induced subgraph  $G \setminus S$  is obtained by removing  $S$  and all edges touching  $S$ .
- Biologically, this models drug or viral inhibition.

## Reachability

- A node is functional only if it receives a signal from an entry node.
- Nodes with no incoming edges cannot transmit biological signals.

# Vertex Cover

## Definition

- A vertex cover is a set  $C \subseteq V$  such that every edge has at least one endpoint in  $C$ .
- That is, for every  $uv \in E$ , either  $u \in C$  or  $v \in C$ .

## Interpretation

- If all vertices in  $C$  are blocked, then no signal can pass along any edge.
- Vertex covers measure how many proteins must be blocked to destroy signaling.

# Minimum Vertex Cover

## Definition

- The minimum vertex cover  $\tau(G)$  is the smallest possible vertex cover.
- It measures the intrinsic control of a network.

## Biological Meaning

- A small  $\tau(G)$  means few proteins can shut down the entire pathway.
- A large  $\tau(G)$  means the pathway is robust to perturbations.

# Algebraic Meaning of Vertex Cover

## Edge Ideals

- For a graph  $G$ , the edge ideal is

$$I(G) = \langle x_u x_v \mid uv \in E \rangle.$$

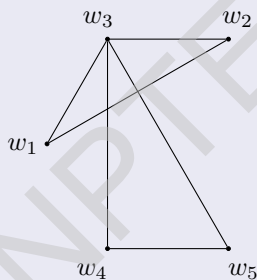
- The height of  $I(G)$  equals the minimum vertex cover  $\tau(G)$ .

## Why This Matters

- Vertex cover is not only combinatorial.
- It is also an algebraic invariant of the signaling network.
- This links biological robustness to algebraic geometry.

# Example

If  $G$  is the following graph:



- A vertex cover (that is not minimal):  $\{w_1, w_2, w_3, w_4\}$
- Minimal Vertex covers:  $\{w_3, w_1, w_4\}$ ,  $\{w_3, w_1, w_5\}$ ,  $\{w_3, w_2, w_4\}$ ,  $\{w_3, w_2, w_5\}$ ,  $\{w_2, w_1, w_4, w_5\}$ .

## Initial JAK/STAT Network

- Source: KEGG pathway hsa04630.
- Proteins (nodes): 155
- Interactions (edges): 868
- Minimum vertex cover: 83

## Entry Nodes

- Nodes 59–99 represent external signal inputs.
- These are never removed during pruning.



# Network Reduction Algorithm

## Drug/Viral Perturbation

- Drugs and viruses delete specific target proteins.
- All incident edges of deleted proteins are removed.

## Iterative Pruning

- Any non-entry node with no incoming edges is removed.
- This deletion propagates until all remaining nodes have incoming signals.
- The remaining graph represents functional signaling.

## Rheumatoid Arthritis Drugs

- Methotrexate: STAT1, STAT5
- Prednisolone: IL2R $\alpha$ , JAK3
- Rituximab: STAT3
- Tocilizumab: IL6 receptor
- Tofacitinib: JAK1, JAK3
- Baricitinib: JAK1, JAK2

## JAK/STAT Inhibition by Viruses

- Coronavirus: IFNAR1, STAT1, STAT2
- Influenza A: IFNAR1, STAT3, IFNGR1, IFNGR2
- Dengue: STAT1, STAT2
- RSV: TYK2, CREB, STAT1, STAT2
- HBV, HCV: JAK1, TYK2, STAT1
- KSV: JAK1, TYK2, STAT2

# Drug Network Sizes

## Single Agents

- Methotrexate: 569 edges, VC 18, ACR20 = 39.7%
- Prednisolone: 527 edges, VC 17, ACR20 = 51%
- Tocilizumab: 589 edges, VC 18, ACR20 = 59%
- Tofacitinib: 473 edges, VC 16, ACR20 = 65.7%
- Baricitinib: 473 edges, VC 16, ACR20 = 77%

# Drug + Methotrexate

## Combination Therapy

- Prednisolone+MTX: 499 edges, VC 17, ACR20 = 70.63%
- Tofacitinib+MTX: 447 edges, VC 16, ACR20 = 75.4%
- Baricitinib+MTX: 447 edges, VC 16, ACR20 = 78%

# Summary of Drugs and Viral Effects (Data Table)

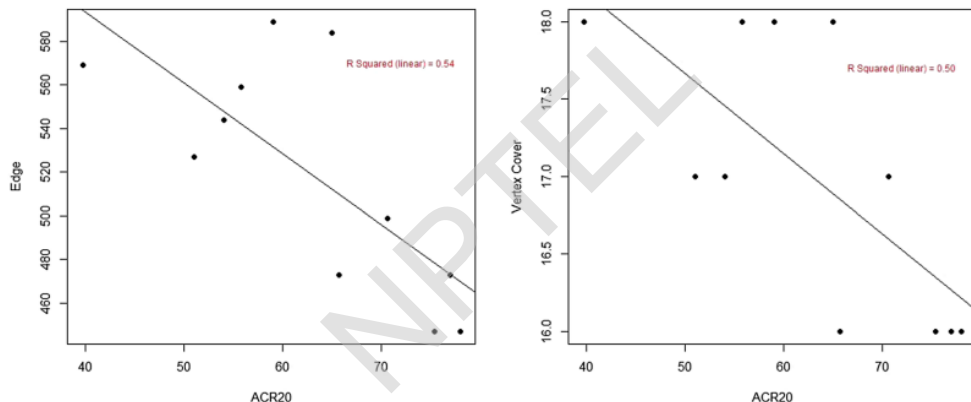
| Drugs used in rheumatoid arthritis |      |              |        |                |      |              |        | Viral systems |      |              |       |              |      |              |         |
|------------------------------------|------|--------------|--------|----------------|------|--------------|--------|---------------|------|--------------|-------|--------------|------|--------------|---------|
| Alone                              | Edge | Vertex cover | ACR 20 | + Methotrexate | Edge | Vertex cover | ACR 20 | Virus         | Edge | Vertex cover | Type  | Virus        | Edge | Vertex cover | Type    |
| Methotrexate                       | 569  | 18           | 39.70  | –              | –    | –            | –      | Measles       | 574  | 18           | Acute | KSV          | 460  | 16           | Chronic |
| Tocilizumab                        | 589  | 18           | 59     | Tocilizumab    | 559  | 18           | 55.80  | Dengue        | 569  | 18           | Acute | HBV          | 460  | 16           | Chronic |
| Ritux                              | 584  | 18           | 65     | Ritux          | 544  | 17           | 54     | Influenza A   | 574  | 18           | Acute | CMV          | 584  | 18           | Acute   |
| Pred                               | 527  | 17           | 51     | Pred           | 499  | 17           | 70.63  | EBV           | 554  | 18           | Acute | Hendra/Nipah | 569  | 18           | Acute   |
| Tofacitinib                        | 473  | 16           | 65.70  | Tofacitinib    | 447  | 16           | 75.40  | RSV           | 508  | 17           | Acute | Corona       | 559  | 18           | Acute   |
| Baricitinib                        | 473  | 16           | 77     | Baricitinib    | 447  | 16           | 78     | WNV/JBV/YFV   | 584  | 18           | Acute | HCV          | 460  | 16           | Chronic |

**Table 1.** Effect of different drugs and viral evasion mechanisms on the network size of the human Janus Kinase/Signal Transducers and Activators of Transcription (JAK/STAT) pathway. ACR 20 responses are expressed as percentages. ACR 20: American College of Rheumatology 20 response; CBV: cytomegalovirus; EBV: Epstein Barr virus; HBV: hepatitis B virus; HCV: hepatitis C virus; JBV: Japanese B virus; KSV: Kaposi's sarcoma virus; RSV: respiratory syncytial virus; WNV: West Nile virus; YFV: yellow fever virus.

## After Excluding Rituximab

- Edge vs ACR20: Spearman  $\rho = -0.815$ ,  $p = 0.007$ ,  $R^2 = 0.676$
- Vertex cover vs ACR20:  $\rho = -0.793$ ,  $p = 0.011$ ,  $R^2 = 0.635$

# Correlation Between Network Size and Clinical Response



**Figure 1.** Represents scatter plot of treatment response (American College of Rheumatology 20 response (ACR 20), x-axis) with edges and minimum vertex cover (y-axes). The data points represent different drugs. The results are obtained with deletion of rituximab from the data set. The linear regression coefficient for edges is 0.676 and that for the minimum vertex cover is 0.635.



# Viral Results

## Acute vs Chronic

- Acute viruses:  $567.7 \pm 21.2$  edges,  $17.91 \pm 0.28$  VC
- Chronic viruses:  $460 \pm 0$  edges,  $16 \pm 0$  VC
- Both differences significant ( $p = 0.002$ )

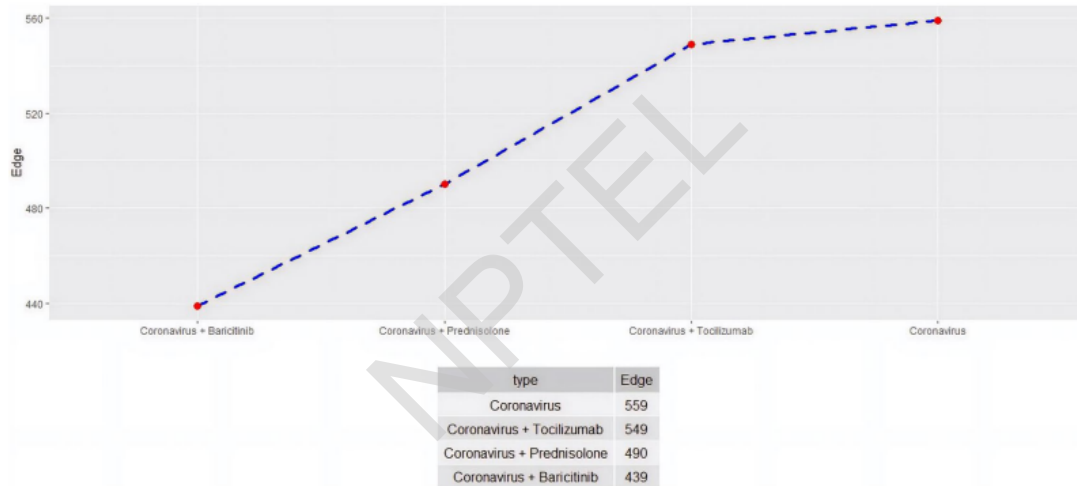
## Coronavirus Alone

- 559 edges, vertex cover 18

## After Drug Addition

- Tocilizumab: minimal change
- Prednisolone: moderate collapse
- Baricitinib: maximum network reduction

# Effect of Immunosuppressive Drugs on COVID-19 Network



**Figure 2.** Network size of the human Janus Kinase/Signal Transducers and Activators of Transcription (JAK/STAT) pathway network infected with coronavirus. The system is further tested in terms of size after introduction of immunosuppressive drugs, namely, prednisolone, tocilizumab and baricitinib. In this figure the number of edges is depicted. Maximum reduction of network edges is obtained with baricitinib, intermediate with glucocorticoids and least with tocilizumab.

## Network Meaning

- Edges measure signal flow capacity.
- Vertex cover measures algebraic control of the pathway.

## Biological Meaning

- Smaller network = weaker immune signaling.
- Stronger drugs collapse the network more.
- Chronic viruses collapse the network more.

# Conclusions

## Main Findings

- JAK/STAT network size predicts drug efficacy.
- Baricitinib is theoretically optimal for COVID cytokine storm.
- Graph invariants link algebra, networks, and immunology.

## References

- 1 A.L. Barabási, N. Gulbahce & J. Loscalzo, Network medicine: a network-based approach to human disease. *Nat Rev Genet* 12, 56–68 (2011): <https://doi.org/10.1038/nrg2918>
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**Module 12: Applications of Random Graphs**  
**Lecture 03: Social Network Analysis**



# Introduction to Social Media Networks

## What is a Social Media Network?

A social media network is a structure where:

- Individuals or accounts are represented as **nodes**.
- Connections (friends, followers) are represented as **edges**.

## Examples

- Facebook (friendship network)
- X : Formerly Twitter (follower network)
- Instagram (follower and interaction network)

# Basic Components of Networks

## Key Elements

- **Nodes:** Users or accounts.
- **Edges:** Relationships (follow, friend, message).
- **Degree:** Number of connections a user has.
- **Path:** Sequence of connections between users.

## Why Important?

- Helps analyze how information travels.
- Identifies important or influential users.

# Types of Social Networks

## Different Network Structures

- **Undirected Networks:** Mutual friendships (Facebook).
- **Directed Networks:** One-way follows (X, formerly Twitter).
- **Weighted Networks:** Strength of interactions (likes, messages).

## Implications

- Structure affects how quickly information spreads.
- Directed networks often create influencers.

# Graph Representation of Social Networks

## Mathematical Model

A social media network is represented as a graph:

$$G = (V, E)$$

where:

- $V$  = set of nodes (users)
- $E$  = set of edges (connections)

## Types of Graphs

- **Undirected Graph:**  $\{u, v\} \in E \Rightarrow \{v, u\} \in E$
- **Directed Graph:**  $(u, v) \neq (v, u)$  (These types of graphs model follower networks)

# Adjacency Matrix Representation

## Adjacency Matrix

A graph  $G = (V, E)$  with  $n$  nodes can be represented by a matrix  $A \in \mathbb{R}^{n \times n}$ :

$$A_{ij} = \begin{cases} 1, & \text{if there is an edge from } i \text{ to } j \\ 0, & \text{otherwise} \end{cases}$$

## Properties

- Undirected graph  $\Rightarrow A$  is symmetric
- Directed graph  $\Rightarrow A$  may not be symmetric

# Degree of a Node

## Degree Definition

For a node  $v_i \in V$ , the degree is:

$$\deg(v_i) = \sum_{j=1}^n A_{ij}$$

## Directed Graph Case

- **In-degree:**  $\deg^-(v_i) = \sum_j A_{ji}$
- **Out-degree:**  $\deg^+(v_i) = \sum_j A_{ij}$

## Interpretation

- Measures how connected a user is in the network

# Paths in Graphs

## Definition of a Path

A path between nodes  $v_1$  and  $v_k$  is a sequence:

$$v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_k$$

such that each consecutive pair is connected by an edge.

## Path Length

- Length = number of edges in the path

## Importance

- Determines how information flows between users

# Weighted Networks

## Weighted Graph

A weighted social network is:

$$G = (V, E, W)$$

where  $W : E \rightarrow \mathbb{R}^+$  assigns weights to edges.

## Adjacency Matrix with Weights

$$A_{ij} = \begin{cases} w_{ij}, & \text{if edge exists} \\ 0, & \text{otherwise} \end{cases}$$

## Interpretation

- $w_{ij}$  represents strength of interaction (likes, messages, etc.)



# Connectivity in Networks

## Connected Graph

A graph  $G = (V, E)$  is connected if:

for all  $u, v \in V$ , there exists a path between  $u$  and  $v$

## Connected Component

A connected component is a maximal subset of nodes where every pair is connected by a path.

## Importance

- Determines whether information can reach all users

# Information Spread in Networks

## How Information Spreads

- A user posts content.
- Their connections see and share it.
- The process continues through the network.

## Key Idea

- Spread depends on connectivity and user activity.
- Highly connected users accelerate diffusion.

# Viral Content and Diffusion

## What is Viral Content?

Content that spreads rapidly to a large number of users.

## Factors Behind Virality

- High number of connections.
- Emotional or engaging content.
- Timing and platform algorithms.

# Role of Influencers

## Who are Influencers?

Users with a large number of followers or strong network positions.

## Impact

- Can spread information quickly.
- Often targeted in marketing campaigns.
- Act as hubs in the network.

## Degree Centrality

Let  $G = (V, E)$  be a graph with vertex set  $V$  and edge set  $E$ .

For a node  $v \in V$ , the degree centrality is defined as:

$$C_D(v) = \deg(v)$$

where  $\deg(v)$  is the number of edges incident to  $v$ .

For normalization:

$$C_D(v) = \frac{\deg(v)}{|V| - 1}$$

## Clustering Coefficient

For a node  $v$  with degree  $k_v$ , the clustering coefficient is:

$$C(v) = \frac{2 \times \text{number of edges between neighbors of } v}{k_v(k_v - 1)}$$

- Measures how close the neighbors of  $v$  are to forming a complete graph.

## Betweenness Centrality

For a node  $v \in V$ , betweenness centrality is defined as:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where:

- $\sigma_{st}$  = number of shortest paths from node  $s$  to node  $t$
- $\sigma_{st}(v)$  = number of those paths passing through  $v$

## Interpretation

- Nodes with high  $C_B(v)$  lie on many shortest paths.
- Such nodes act as **bridges** controlling information flow in the graph.

# Definitions of Network Metrics

## Degree Centrality

Degree centrality of a node is the number of direct connections it has.

- Measures how **popular or active** a user is.
- Higher degree means more immediate influence.

## Clustering Coefficient

The clustering coefficient measures how connected a node's neighbors are to each other.

- Indicates the presence of **tight-knit groups**.
- High value means friends of a user are also friends with each other.

## Betweenness Centrality

Betweenness centrality measures how often a node lies on the shortest paths between other nodes.

- Captures a node's role as a **bridge or connector**.
- High betweenness means strong control over information flow.

## Summary

- Degree → popularity
- Clustering → community structure
- Betweenness → control of information flow



# Network Metrics in Social Media

## Important Metrics

- **Degree Centrality:** Number of connections.
- **Clustering Coefficient:** How connected neighbors are.
- **Betweenness Centrality:** Control over information flow.

## Use

- Identify key users.
- Understand community structure.

# Communities in Social Networks

## What are Communities?

Groups of users who are more connected to each other than to the rest of the network.

## Importance

- Helps target specific audiences.
- Explains echo chambers and group behavior.

# Challenges in Social Media Networks

## Spread of Misinformation

- Social media platforms allow information to propagate rapidly, which can lead to the widespread dissemination of false or misleading content.
- Misinformation can influence public opinion, behaviors, and even election outcomes.

## Privacy Concerns and Data Misuse

- Users often share personal data unknowingly, which can be exploited by advertisers, political actors, or hackers.
- Data misuse can result in identity theft or unwanted surveillance.

# Challenges in Social Media Networks

## Network Bias and Echo Chambers

- Algorithms often show content similar to users' existing beliefs, creating isolated communities.
- This leads to filter bubbles and reinforcement of existing biases, limiting exposure to diverse viewpoints.

## Cyberbullying and Political Manipulation

- Social networks can facilitate harassment, trolling, or bullying, impacting victims' mental health.
- Platforms can be used to spread propaganda or manipulate public opinion, sometimes affecting elections.

# Why Study Social Networks

## Policy and Ethical Guidelines

- Understanding network dynamics helps design regulations to protect users.
- Policies can mitigate harmful behavior and promote safe online interactions.

## Content Control and Healthy Discourse

- Analyzing network propagation patterns allows for controlling misinformation or harmful content.
- Insights help promote constructive conversations and reduce online polarization.

# Applications of Network Analysis

## Marketing and Public Health

- Companies use network analysis to identify influential users and deliver personalized marketing campaigns.
- Public health agencies track disease spread and target awareness campaigns using social network insights.

## Recommendation Systems and Fraud Detection

- E-commerce and streaming platforms suggest items based on social connections and similar user behavior.
- Network patterns can reveal suspicious activity, fake accounts, or financial fraud.

# Applications of Network Analysis

## Political and Organizational Use

- Political campaigns use network insights to identify opinion leaders and understand message propagation.
- Businesses optimize supply chains and collaboration networks using network analysis.

# Impact on Society

## Positive Impacts

- Enables rapid sharing of information, educational resources, and emergency alerts.
- Facilitates collaboration and community building among people with common interests.
- Supports social movements and awareness campaigns.

## Negative Impacts

- Amplifies misinformation and fake news.
- Creates echo chambers that reinforce biases.
- Can lead to harassment, radicalization, or exposure to harmful content.



# Challenges in Data Collection

## Issues and Solutions

- Large datasets are computationally intensive and may contain noise or missing values.
- Privacy laws such as GDPR restrict access to data, requiring ethical collection and anonymization.
- Sampling and advanced algorithms help ensure meaningful and reliable analysis.

# Future Directions in Network Analysis

## Emerging Trends

- AI and machine learning for predicting behavior, content spread, and network evolution.
- Multilayer networks to analyze multiple types of interactions simultaneously.
- Real-time monitoring to detect viral content, misinformation, and crises.
- Personalized recommendations while minimizing filter bubbles.
- Designing interventions to curb misinformation and promote positive behaviors.

# Conclusion

## Summary

- Social media platforms can be analyzed as networks.
- Connections determine how information spreads.
- Mathematical models help quantify and predict behavior.

Understanding social media networks enables better communication, marketing, and policy decisions in the digital world.

## References

- 1 A. Marin, & B. Wellman (2011). Social network analysis: An introduction. The SAGE handbook of social network analysis, 11, 25.
- 2 S. Wasserman and K. Faust (2012): Social Network Analysis: Methods and Applications. Cambridge: Cambridge University Press.

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**Module 12: Applications of Random Graphs**

**Lecture 04: Simple Linear Regression on Social  
Network Data**



# Social Media Networks

## Understanding Social Media

Social media platforms such as Facebook, X (formerly Twitter), and Instagram can be viewed as networks where:

- Users are **nodes (vertices)**.
- Friendships, follows, or connections are **edges**.

Information spreads through these connections in the form of posts, messages, or shares.

## Why This Matters

- Explains how viral content spreads across users.
- Helps platforms design better recommendation algorithms.
- Important for understanding influence and reach in online communities.

# Applications in Social Media

## Key Applications

- **Viral Marketing:** Brands use highly connected users to promote products.
- **Influencer Strategy:** Identifying users with large reach for advertising.
- **Public Awareness:** Spreading important messages (e.g., health campaigns).

## Insights from the Study

- Even users with fewer connections can reach many people due to network structure.
- Increasing connections helps, but **quality and position in the network also matter**.
- Helps in designing efficient strategies for maximizing information spread.



# Network Data Analysis

In network analysis, we often want to quantify the effect of a network property on a network outcome.

- **Predictor ( $X$ ):** a network metric such as:
  - **Number of edges:** total edges in the network
  - **Average degree:** mean connections per node
  - **Vertex cover size:** minimum set of nodes that touches all edges
- **Response ( $Y$ ):** a network outcome such as:
  - **Robustness:** fraction of nodes in the largest connected component after random node removal
  - **Diffusion efficiency:** proportion of nodes reached by a spreading process
  - **Network connectivity:** fraction of nodes that remain reachable
- **Error term ( $\varepsilon_i$ ):** random variability not explained by  $X$

# Real-Life Meaning of the Model

## Connecting Model to Reality

- **Predictor ( $X$ ):** Number of friends/followers a user (or network) has.
- **Response ( $Y$ ):** How many people actually see or receive the message.

The regression model helps us quantify:

Does having more connections actually increase reach?

## Practical Interpretation

- A higher number of connections generally increases visibility.
- However, the increase may be **small per additional connection**.
- Not all friendships are equally influential in spreading information.

# Why This Study Matters

## Motivation

Understanding how friendships influence information spread helps us design better communication strategies in social networks, marketing campaigns, and public awareness initiatives.

## Importance of the Study

- Helps identify whether increasing connections improves message reach.
- Provides insights for optimizing network structures for faster information flow.
- Useful for applications such as viral marketing, rumor control, and emergency communication.

## Simple Linear Regression Model

We assume a linear relationship between predictor  $X$  and response  $Y$ :

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, 2, \dots, n$$

- $\beta_0$  (intercept): expected outcome when  $X = 0$
- $\beta_1$  (slope): expected change in  $Y$  for a one-unit increase in  $X$
- $\varepsilon_i$ : residual variability around the regression line

# Estimation of Parameters

## Least Squares Estimation

We estimate  $\beta_0$  and  $\beta_1$  by minimizing the sum of squared differences between observed and predicted values:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

The closed-form solutions are:

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

## Interpretation

- $\hat{\beta}_1$  tells us the expected change in the network outcome  $Y$  for a one-unit change in the network metric  $X$ .
- $\hat{\beta}_0$  represents the expected outcome when the network metric is zero.
- Fitted line:  $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$

# Assumptions of Linear Regression

## Key Assumptions

For the Simple Linear Regression model to provide valid inference, these assumptions should hold:

- **Linearity:** The expected outcome changes linearly with the network metric.
- **Independence:** Observations (different networks or measurements) are independent.
- **Homoscedasticity:** The variability of  $Y$  around the regression line is constant across all values of  $X$ .
- **Normality:** The residuals  $\varepsilon_i$  are normally distributed (important for confidence intervals and hypothesis testing).

## Context to Social Network

- Linearity: more edges gives higher diffusion; small networks may deviate slightly.
- Independence: each network generated or measured independently.
- Homoscedasticity: variability in robustness should not increase with network size.
- Normality: random fluctuations in network outcomes approximate normal noise.

# Example: Context and Objective

## Study Objective

We want to understand how the number of friendships in a social network affects the efficiency of information spread.

## Key Questions

- Does having more friendships help messages reach more people?
- Can we quantify the relationship using a simple regression model?

# Simulated Network Setup

## Network Simulation Details

- Simulate 30 social networks with 50 people each.
- Networks generated using Erdős–Rényi model  $G(n = 50, p_i)$  with varying  $p_i$  to simulate sparse and dense networks.
- Predictor ( $X$ ): total number of friendships (edges).
- Response ( $Y$ ): fraction of people reached by a message starting from 5 randomly informed people.
- Gaussian noise added to simulate natural variability in information spread.

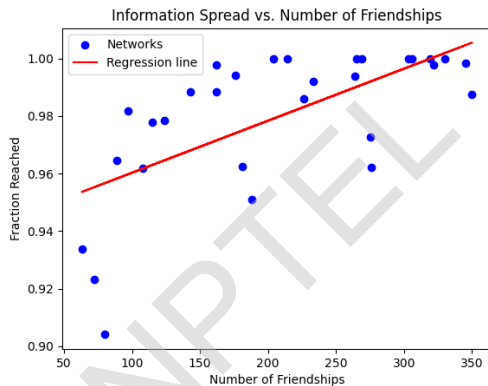


# Simulating Information Spread

## Spread Process

- Start with 5 randomly informed nodes.
- At each step, newly informed nodes inform all their neighbors.
- Process continues until no new nodes are informed.
- Fraction reached  $Y_i = \frac{\text{Number of informed nodes}}{\text{Total nodes}}$ .
- Noise ensures realistic variability between similar networks.

# Scatter Plot and Regression



## Visualizing the Relationship

- Each point represents one simulated network.
- X-axis: total number of friendships (edges).
- Y-axis: fraction of people reached.
- Red line: fitted regression  $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ .
- Observation: networks with more friendships tend to reach a larger fraction of people.

# Regression Model

## Simple Linear Regression

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- $Y_i$ : fraction of people reached (information spread)
- $X_i$ : number of friendships
- $\beta_0$ : baseline spread when  $X = 0$
- $\beta_1$ : change in spread per additional friendship
- $\varepsilon_i$ : residual (unexplained variability)

# Estimated Coefficients

## Regression Results from Simulation

| Coefficient     | Estimate | Std. Error |
|-----------------|----------|------------|
| $\hat{\beta}_0$ | 0.942221 | 0.009003   |
| $\hat{\beta}_1$ | 0.000181 | 0.000040   |

## Interpretation

- Baseline spread when there are 0 friendships:  $\hat{\beta}_0 = 0.942$  ( approximately 94% of nodes reached from random seeding).
- Each additional friendship increases the fraction reached by  $\hat{\beta}_1 = 0.00018$  (a very small increment per friendship).
- The slope is positive, confirming that more connected networks slightly improve information spread efficiency.
- High baseline indicates that even sparse networks already allow a large fraction of nodes to be reached due to network connectivity.

# Small Network Example

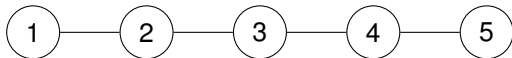
## Small Networks with 5 Nodes Each

We illustrate how network structure affects information spread.

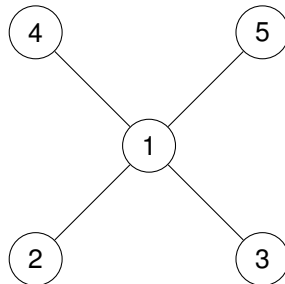
- Graph 1: Chain structure
- Graph 2: Star structure
- Graph 3: Cycle structure
- Graph 4: Sparse random connections

## Small Network Examples

Graph 1: Chain (Edges = 4)

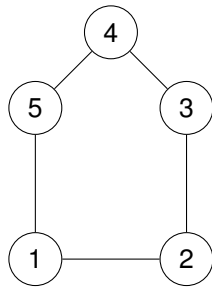


Graph 2: Star (Edges = 4)

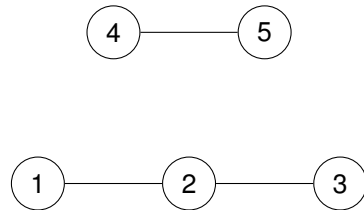


## Small Network Examples

Graph 3: Cycle (Edges = 5)



Graph 4: Sparse Random (Edges = 3)



# Response in Network Spread

## Definition of Response $Y$

In our network model, the response variable  $Y$  is defined as the fraction of nodes reached by information starting from initially informed nodes:

- Let  $V$  be the set of all nodes in the network, with  $|V| = n$ .
- Let  $I_0 \subseteq V$  be the set of initially informed nodes (for our example,  $|I_0| = 1$ ).
- Let  $I_\infty \subseteq V$  be the set of all nodes that eventually receive the information.
- Then the response  $Y$  is:

$$Y = \frac{|I_\infty|}{|V|}$$

- Example 1: In a 5-node chain starting from node 1:  $I_\infty = \{1, 2, 3, 4, 5\} \Rightarrow Y = 1$ .
- Example 2: In a sparse disconnected graph:  $I_\infty = \{1, 2, 3\} \Rightarrow Y = 0.6$ .



# Predictor and Response

## Regression Setup

We compute a simple linear regression to see how network connectivity affects spread.

- Predictor  $X$  = number of edges (friendships)
- Response  $Y$  = fraction of nodes reached starting from 1 node

| Graph     | X (Edges) | Y (Fraction Reached) |
|-----------|-----------|----------------------|
| G1 Chain  | 4         | 1.0                  |
| G2 Star   | 4         | 1.0                  |
| G3 Cycle  | 5         | 1.0                  |
| G4 Sparse | 3         | 0.6                  |

# Regression Analysis

## Simple Linear Regression

We fit the model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

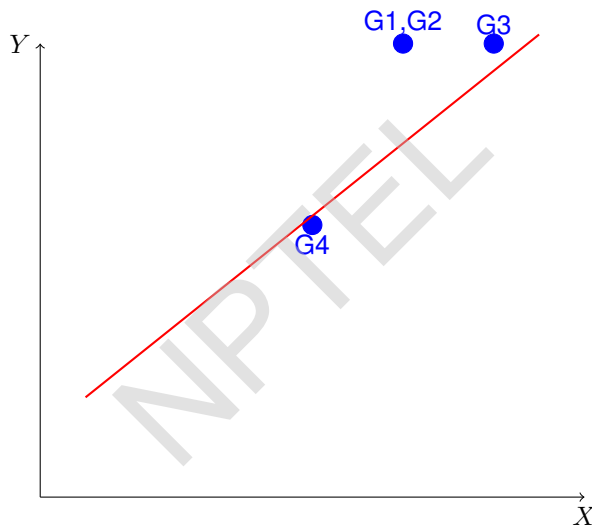
- $\hat{\beta}_1 = 0.2$  (increase in spread per additional edge)
- $\hat{\beta}_0 = 0.1$  (baseline fraction reached with 0 edges)

## Regression Line

$$\hat{Y} = 0.1 + 0.2X$$

- Sparse network ( $X=3$ )  $\rightarrow \hat{Y} = 0.7$ , close to 0.6 observed
- Dense networks ( $X=4$  or  $5$ )  $\rightarrow$  predicted  $Y = 1$  (capped at 1)

## Regression Plot (Conceptual)



### Interpretation

Networks with more edges generally reach more nodes. Sparse networks have lower spread. The regression line captures this trend.

# Conclusions

## Key Findings

- Networks have a high baseline information spread ( $\approx 94\%$ ) even with few edges.
- Additional friendships have a small but positive effect on increasing spread efficiency.
- The relationship between friendships and spread is approximately linear in this regime.
- Noise introduces realistic variability between networks but does not obscure the positive trend.

## Implications

- Even sparse networks are fairly effective for information diffusion.
- Adding connections improves spread only incrementally—dense networks are not strictly necessary for near-complete coverage.
- Useful for network design, viral marketing, and studying diffusion dynamics in social networks.

## References

- 1 A. Marin, & B. Wellman (2011). Social network analysis: An introduction. The SAGE handbook of social network analysis, 11, 25.
- 2 S. Wasserman and K. Faust (2012): Social Network Analysis: Methods and Applications. Cambridge: Cambridge University Press.

*Thank you*





# SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 12: Applications of Random Graphs

Lecture 05: Conclusion and Course Summary



# Course Overview and Conclusions

## Motivation

- Modern scientific, biological, and social systems generate massive, complex, and high-dimensional data.
- Classical statistical tools alone are insufficient to analyze such data effectively.
- Big data analysis requires a synthesis of statistics, linear algebra, optimization, and high-performance computing.

## What This Course Achieves

- Builds statistical foundations for analyzing high-dimensional data.
- Connects classical inference with modern statistical learning theory.
- Develops multivariate and network-based modeling tools.
- Emphasizes real-world applications and case studies.

## Key Takeaway

- Understanding structure, dimension, and dependence is essential.
- Graphs, multivariate models, and learning theory provide unifying principles.



## Introduction to Big Data

- Definition and characteristics of big data.
- Examples and case studies illustrating big data challenges.
- Difference between big data analysis and big data processing.
- Role of high-dimensional statistics in big data analysis.
- Classical statistical inference vs big data inference.
- Importance of linear algebra, calculus, and high-performance computing.

### Basic Statistical Inference I: Point Estimation

- Concept of point estimation.
- Unbiased estimation.
- Maximum Likelihood Estimation (MLE).
- Method of Moments Estimation.
- Asymptotic properties of estimators.
- Case studies and applications.

### Basic Statistical Inference II

- Concept of interval estimation.
- Confidence intervals and interpretation.
- Fundamentals of hypothesis testing.
- Test statistics, p-values, and decision rules.
- Illustrative examples and applications.

### Statistical Learning Theory I

- Bias and variance of estimators.
- Mean Squared Error (MSE).
- Relationship between model complexity and prediction accuracy.
- Real-life examples and case studies.

### Statistical Learning Theory II

- Bias–variance trade-off.
- Underfitting and overfitting.
- Comparison of competing estimation models.
- Case studies demonstrating model selection.

### Multivariate Data and Linear Models

- Multivariate data structures.
- Simple and multiple linear regression.
- Multivariate regression models.
- Gauss–Markov Theorem.
- Applications in real data analysis.

### Multivariate Probability Distributions

- Introduction to multivariate distributions.
- Multivariate normal distribution.
- PDF, CDF, and normality.
- Conditional distributions.
- Linear and affine transformations.
- Other common multivariate distributions.

### Multivariate Analysis and Dimensionality

- Concept of unsupervised learning.
- Hierarchical and non-hierarchical clustering.
- Detailed study of K-means clustering.
- Curse of dimensionality.
- Asymptotic behavior of volume and clustering performance.



### Population Principal Component Analysis

- Concept and motivation of PCA.
- Population covariance structure.
- Eigenvalues and eigenvectors.
- Dimensionality reduction.
- Applications and case studies.

### Sample Principal Component Analysis

- Sample covariance matrix.
- Sample eigenvalues and eigenvectors.
- Differences between population and sample PCA.
- High-dimensional effects.
- Applications and case studies.

## Network Data Analysis

- Introduction to network data.
- Graph representations.
- Random graph models.
- Laws of large numbers for random graphs.
- Properties of large-scale networks.

## Course Outline: Week 12

### Applications of Random Graphs

- Network Medicine.
- Applications of Random Graphs to Social Networks.
- Course summary and integration of concepts.

*Thank you*

